Light

GCSE Computer Science - Embedded Systems (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement correctly describes an embedded system?

- ○ **A.** A computer that can be taken apart easily
- ○ **B.** Any computer that uses electricity
- ○ **C.** A computer system integrated into another device to control specific functions
- ○ **D.** A computer that only works when connected to the internet

Q2 (1 mark)

The software in most embedded systems is:

- ○ **A.** Changed by the user every day
- ○ **B.** Fixed at manufacture and designed for specific tasks
- ○ **C.** Downloaded from app stores
- ○ **D.** The same as desktop operating systems

Q3 (1 mark)

A dishwasher contains a controller that manages water temperature, wash cycles, and timing. This controller is:

- ○ **A.** A desktop computer
- ○ **B.** A games console
- ○ **C.** A cloud server
- ○ **D.** An embedded system

Q4 (1 mark)

A car's parking sensor system detects obstacles and warns the driver with beeping sounds. What type of system controls this?

- ○ **A.** A word processor
- ○ **B.** An embedded system
- ○ **C.** A spreadsheet application
- ○ **D.** A social media platform

Q5 (1 mark)

Embedded systems typically start working immediately when powered on. Why is this important?

- ○ **A.** It is not important - users prefer waiting
- ○ **B.** Embedded systems have very slow processors
- ○ **C.** Many embedded systems control time-critical functions where delays could cause problems or safety issues
- ○ **D.** Users must configure the system each time it starts

Q6 (1 mark)

Many embedded systems must operate reliably in harsh environments. Which is an example of this?

- ○ **A.** An engine controller working at high temperatures with vibration
- ○ **B.** A laptop used in an air-conditioned office
- ○ **C.** A desktop computer on a stable desk
- ○ **D.** A tablet used indoors for reading

Q7 (1 mark)

An electric toothbrush with a timer and pressure sensor contains:

- ○ **A.** A mainframe computer
- ○ **B.** An embedded system

- ○ **C.** A network server
- ○ **D.** A virtual machine

Q8 (1 mark)

Embedded systems are often designed to operate for years with minimal maintenance. Why is this necessary?

- ○ **A.** Many embedded systems are in locations that are difficult to access or where continuous operation is critical
- ○ **B.** Maintenance is always unnecessary for any computer
- ○ **C.** Embedded systems never break down
- ○ **D.** Users enjoy performing frequent maintenance

Q9 (1 mark)

A CCTV security camera that records video and detects motion contains:

- ○ **A.** Only a glass lens with no electronics
- ○ **B.** An embedded system for video processing and motion detection
- ○ **C.** A full desktop computer inside the camera housing
- ○ **D.** No processing capability at all

Q10 (1 mark)

Because embedded systems perform dedicated functions, their hardware can be:

- ○ **A.** Exactly the same as desktop computers
- ○ **B.** More expensive than supercomputers
- ○ **C.** Optimised specifically for those functions, improving efficiency
- ○ **D.** Unable to perform any useful work

Q11 (1 mark)

An automatic hand dryer in a public toilet detects hands and activates the heating element and fan. This is controlled by:

- ○ **A.** A human operator
- ○ **B.** An embedded system
- ○ **C.** A desktop PC hidden in the wall
- ○ **D.** Cloud computing

Q12 (1 mark)

Compared to general-purpose computers, embedded systems typically offer:

- ○ **A.** More flexibility but lower reliability
- ○ **B.** Larger physical size and higher power consumption
- ○ **C.** The ability to run any software the user chooses

- ○ **D.** Less flexibility but higher reliability for their specific function

Q13 (2 marks)

A coffee machine with programmable settings brews different types of coffee automatically.
Describe two functions the embedded system performs.

- ○ **A.** Playing music and displaying advertisements
- ○ **B.** Connecting to social media and sending emails
- ○ **C.** Controlling water temperature precisely and managing brewing time for different coffee types
- ○ **D.** Running office productivity software

Q14 (2 marks)

Embedded systems typically have less memory (RAM) than general-purpose computers.
Explain why this is usually sufficient.

- ○ **A.** Embedded systems do not use memory at all
- ○ **B.** Users add more memory themselves when needed
- ○ **C.** Memory is too expensive for any embedded system
- ○ **D.** Dedicated software performing specific tasks requires less memory than general-purpose operating systems running multiple applications

Q15 (2 marks)

A cardiac pacemaker regulates heartbeat using electrical impulses. Explain why embedded system characteristics are essential for this application.

- ○ **A.** Pacemakers need to play music to keep patients entertained
- ○ **B.** Pacemakers require high-resolution displays
- ○ **C.** Extreme reliability is essential as failures could be fatal; small size enables implantation; low power allows years of battery life inside the body
- ○ **D.** Patients need to install their own applications on pacemakers

Q16 (2 marks)

Traffic light controllers manage signal timing at road junctions. Identify two requirements that make embedded systems suitable.

- ○ **A.** Need for word processing and email features
- ○ **B.** Ability to browse social media websites
- ○ **C.** Requirement for high-definition gaming graphics
- ○ **D.** Continuous 24/7 operation without crashes and real-time response to sensor inputs detecting vehicles

Q17 (2 marks)

Many embedded systems store their programs in ROM (Read-Only Memory) or flash memory.
Explain two advantages of this approach.

- ○ **A.** Programs are retained when power is removed, and they are available immediately at startup without loading delays

- ○ **B.** Programs can be changed easily by any user
- ○ **C.** ROM requires constant power to maintain data
- ○ **D.** Programs must be downloaded from the internet each time

Q18 (2 marks)

A home broadband router manages internet connections for multiple devices.
Describe two functions its embedded system performs.

- ○ **A.** Playing video games and streaming movies for users
- ○ **B.** Routing network packets between devices and the internet, and managing wireless connections with security encryption
- ○ **C.** Editing photographs and creating documents
- ○ **D.** Running desktop operating system applications

Q19 (2 marks)

Embedded systems process input signals and produce output signals. Give an example of each for a room thermostat.

- ○ **A.** Input: keyboard typing; Output: printed document
- ○ **B.** Input: music from streaming service; Output: video display
- ○ **C.** Input: temperature reading from sensor; Output: signal to heating system to turn on or off
- ○ **D.** Input: email messages; Output: social media posts

Q20 (2 marks)

An anti-lock braking system (ABS) prevents wheels from locking during hard braking.
Why must this embedded system respond extremely quickly?

- ○ **A.** Wheel lock-up develops in milliseconds, and delayed intervention would result in loss of control - real-time response is essential for safety
- ○ **B.** Quick response is not important for braking systems
- ○ **C.** ABS systems only operate when the car is stationary
- ○ **D.** Slow response would make the ride more comfortable

Q21 (2 marks)

Embedded systems in safety-critical applications undergo extensive testing.
Explain why this level of testing is justified.

- ○ **A.** Testing is optional and usually skipped
- ○ **B.** Users prefer to find bugs themselves
- ○ **C.** Post-deployment updates are often impractical, and failures could cause injury or death - thorough pre-release testing is essential
- ○ **D.** Testing makes products more expensive so should be minimised

Q22 (2 marks)

An air conditioning system monitors room temperature and adjusts cooling output.
Identify two embedded system characteristics this requires.

- ○ **A.** Ability to play video games and browse websites
- ○ **B.** Keyboard and mouse for user interaction
- ○ **C.** High-resolution display for watching films
- ○ **D.** Continuous sensor monitoring throughout operation and reliable control of compressor and fan motors

Q23 (2 marks)

Embedded systems often have longer operational lifespans than general-purpose computers.
Explain why this is the case.

- ○ **A.** Dedicated functions do not become obsolete, there is no need for hardware upgrades to run new software, and replacement is often costly or impractical
- ○ **B.** Embedded systems are less reliable so fail more often
- ○ **C.** Users constantly want to upgrade to newer models
- ○ **D.** Embedded systems are always replaced annually

Q24 (3 marks)

A drone's flight controller maintains stable flight by processing sensor data and adjusting motor speeds.
Analyse the embedded system requirements for this application.

- ○ **A.** The drone only needs to look attractive without electronic control
- ○ **B.** Flight control can wait several seconds before responding to changes

○ **C.** Drones do not require any processing capability

○ **D.** Real-time processing of accelerometer and gyroscope data enables immediate motor adjustments; processing loops run hundreds of times per second to maintain stability; failure would cause immediate crash

Q25 (3 marks)

An automatic pet feeder dispenses food at scheduled times and can be controlled via smartphone app.
Analyse how embedded and connected features work together in this device.

○ **A.** The embedded controller manages local scheduling and motor control for dispensing; WiFi connectivity enables remote monitoring and schedule changes; local operation continues if network fails

○ **B.** The feeder uses no technology and requires manual operation

○ **C.** Only cloud servers control the feeder with no local processing

○ **D.** The feeder requires constant internet connection to dispense any food

Q26 (3 marks)

Battery-powered embedded systems use sleep modes to extend battery life.
Analyse how sleep modes work and when they are used.

○ **A.** Sleep modes use more power than full operation

○ **B.** The processor enters low-power states when not actively processing, waking periodically or on events; this reduces average power consumption significantly while maintaining responsiveness when needed

- ○ **C.** Sleep modes completely shut down the device permanently
- ○ **D.** Battery-powered devices must always run at full power

Q27 (3 marks)

A laser printer contains embedded systems controlling paper handling, laser imaging, and fusing.
Analyse why multiple embedded components might be used rather than a single controller.

- ○ **A.** Using multiple components is purely to increase manufacturing cost
- ○ **B.** Printers do not require any electronic control systems
- ○ **C.** A single controller would always be more reliable
- ○ **D.** Different subsystems have different timing requirements and can be optimised independently; parallel processing handles simultaneous tasks; failure in one subsystem does not crash the entire printer

Q28 (3 marks)

Connected embedded systems face security threats that isolated systems did not.
Analyse the security challenges and how they might be addressed.

- ○ **A.** Network connectivity exposes systems to remote attacks; threats include unauthorised access, malware, and data theft; protection requires encryption, authentication, secure boot, and regular security updates
- ○ **B.** Connected embedded systems face no security risks
- ○ **C.** Security only matters for desktop computers, not embedded systems

○ **D.** Disconnecting from all networks solves all security problems with no downsides

Q29 (3 marks)

A car's GPS navigation system provides directions using satellite signals and digital maps.
Analyse the embedded system requirements for real-time navigation.

○ **A.** GPS navigation requires no processing - it just displays a static map

○ **B.** Navigation systems only work when the car is stationary

○ **C.** The system must process satellite signals to calculate position, correlate location with map data, calculate routes in real-time, and provide timely turn-by-turn guidance that updates as the vehicle moves

○ **D.** Paper maps are built into all GPS units

Q30 (3 marks)

Embedded systems continue to evolve with technological advances. Analyse how trends like artificial intelligence might affect embedded systems development.

○ **A.** AI enables local decision-making without cloud connectivity; edge processing reduces latency for time-critical applications; challenges include power consumption and processing requirements of AI algorithms on constrained hardware

○ **B.** Embedded systems will remain unchanged regardless of any technological advances

○ **C.** Artificial intelligence has no applications in embedded systems

○ **D.** All embedded systems will be replaced by smartphones

Total: **55** marks • Time: **30 min**

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